QBank Quiz December 24, 2024

Test ID: 322120050

Question #1 of 10

Question ID: 1704767

You receive the following error message after addressing and enabling an interface:

`%192.168.16.0 overlaps with FastEthernet0/0`

Which of the following are NOT the causes of the error message? (Choose two.)

A) Incorrect subnet mask in the new interface

B) **Incorrect encapsulation configured**

C) **Failure to issue the no shutdown command**

D) Incorrect IP address on the new interface

Explanation

The error message `%192.168.16.0 overlaps with FastEthernet0/0` indicates that the newly configured interface is in the same subnet as an existing interface. This can occur if there is an incorrect subnet mask or an address that inadvertently places the new interface in that subnet. Each router interface must be in a different subnet to function. For example, when the series of commands below is executed on a router, it will elicit the error message because the two IP addresses used are in the same subnet given the subnet mask in use.

```
Router#config t
Router(config)#interface S0
Router(config-if)#ip address 192.168.1.17 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)interface S1
Router(config-if)#ip address 192.168.1.65 255.255.255.0
Router(config-if)#no shutdown
0%192.168.1.0 overlaps with Serial 0
```

An incorrect encapsulation would prevent the interface from working but would not generate this message.

If the **no shutdown** command had not been issued, we would not be receiving this error. It is only generated when an attempt is made to enable an incorrectly configured interface.

Objective:
Network Fundamentals

Sub-Objective:

Identify interface and cable issues (collisions, errors, mismatch duplex, and/or speed)

References:

Cisco > Support > IP Addressing: IPv4 Addressing Configuration Guide, Cisco IOS XE Release 3S > Chapter: Configuring IPv4 Addresses

Cisco > Support > Basic Router Configuration

Question #2 of 10

Question ID: 1703554

Which of the following IP addresses are valid Class B host addresses if a default Class B mask is in use? (Choose all that apply.)

A) 10.6.8.35

B) 127.0.0.1

C) 190.6.5.4

D) 133.6.5.4

E) **192.168.5.9**

Explanation

The IP addresses 133.6.5.4 and 190.6.5.4 are both valid Class B addresses when a default mask is in use. The Class B default mask is 255.255.0.0 and the range of valid addresses is 128.0.0.0 – 191.255.255.255.

The IP address 10.6.8.35 is a Class A address. The Class A default mask is 255.0.0.0 and the range of valid addresses is 1.0.0.0 – 127.255.255.255, with the exception of the range 127.0.0.1 – 127.255.255.255, which is reserved and cannot be assigned.

The IP address 192.168.5.9 is a Class C address. The Class C default mask is 255.255.255.0 and the range of valid addresses is 192.0.0.0 – 223.255.255.255.

The IP address 127.0.0.1 is a Class A address, but it comes from a reserved portion that cannot be assigned. The range 127.0.0.1 – 127.255.255.255 is used for diagnostics, and although any address in the range will work as a diagnostic address, 127.0.0.1 is known as the loopback address. If you can ping this address, or any address in the 127.0.0.1 – 127.255.255.255 range, then the NIC is working, and TCP/IP is installed.

Objective:

Network Fundamentals

Sub-Objective:

Configure and verify IPv4 addressing and subnetting

References:

Cisco > Support > Technology Support > IP Routing > Troubleshooting TechNotes > Configure IP Addresses and Unique Subnets for New Users > Document ID: 13788

Question #3 of 10

Question ID: 1703583

You have discovered that hackers are gaining access to your WEP wireless network. After researching, you discover that the hackers are using war-driving methods. You need to protect against this type of attack.

What should you do? (Choose all that apply.)

A) Change the default SSID.

B) **Configure the network to use authenticated access only.**

C) **Configure the WEP protocol to use a 128-bit key.**

D) Disable SSID broadcast.

Explanation

You should complete the following steps to protect against war-driving attacks:

- Change the default SSID.
- Disable SSID broadcast.
- Configure the network to use authenticated access only.

You should not configure the WEP protocol to use a 128-bit key. In recent years, WEP has been proven to be an ineffective security protocol for wireless networks, regardless of whether you use low or high encryption. It is a protocol that is very easy to crack using a brute force attack. Some other suggested steps include the following:

- Implement Wi-Fi Protected Access (WPA), WPA2, or WPA3 instead of WEP, preferably WPA3.
- Reduce the access point's signal strength.
- Use MAC filtering.

War driving is a method of discovering 802.11 wireless networks by driving around and looking for open wireless networks. NetStumbler is a common war-driving tool.

For the Cisco CCNA exam, you need to understand the following wireless security concepts and how to harden the wireless network: WEP, WPA/WPA2/WPA3, TKIP, AES, 802.1x, TLS/TTLS, and MAC filtering.

- WEP – You should avoid this protocol because it is easy to crack. Instead, you should use WPA WPA2 or WPA3. WEP uses a 64-bit (low encryption mode) or 128-bit (high encryption mode) key.
- WPA/WPA2/WPA3:
  - WPA uses Temporal Key Integrity Protocol (TKIP) while WPA2 is capable of using TKIP or the more Advanced Encryption Standard (AES) algorithm. WPA was created as a more secure alternative to WEP.
  - WPA2 is stronger than WPA but requires more processing power. Use WPA2 if all the devices are capable of it because it provides protection only exceeded by WPA3.. The only time that using WPA would be sufficient is when your access point or other devices are not capable of supporting WPA2.
    - Enterprise – This version of WPA2 uses security certificates and requires the use of a Remote Authentication Dial-In User Service (RADIUS) authentication server.
    - Personal – This version of WPA2 provides adequate protection for a small office or home network.

- WPA3 is more secure than WPA or WPA2. While WPA2 was a significant increase in security over WPA, WPA3 makes additional improvements over WPA2, including:
- WPA3-Personal, unlike WPA2-Personal which uses pre-shared keys (passphrases), utilizes a process called Simultaneous Authentication of Equals (SAE) in which the station and access point authenticate one another by proving to each other that they have a key. This produces a master key (PMK) which, while shared between the two, is never sent. So, should a hacker capture the packets used, the original key or the generated master key cannot be derived.

- WPA3-Enterprise includes a new 192-bit security level based on the NSA's requirements for environments requiring greater security. It also only allows GCMP-256 encryption.
- 802.1x is an authentication method that can be used on both wired and wireless LANS. An 802.1x client attaches to an 802.1x supplicant (a wireless router or switch), which then forwards the request to an authenticator (a RADIUS server).
- Transport Layer Security (TLS) and Tunneled Transport Layer Security (TTLS) can be implemented with EAP. TLS requires a client certificate, while TTLS does not. TTLS uses a tunnel to connect the server to the client.
- MAC filtering allows or denies a wireless connection based on the client's media access control (MAC) address. The most popular form of MAC filtering will only allow connections to devices that are contained in the list. In this case, you must add any new devices specifically to the MAC filter list on the access point to allow the devices to connect. In contrast, you can configure a deny list, which specifically denies connections to devices with the MAC addresses listed. This form of MAC filtering is not used as much.

Let us look at an example regarding MAC filtering. Suppose encrypted wireless access points are used at a retail location for inventory and price verification. If the retail location is located in a mall, unauthorized access to the wireless network could be a constant problem. If a finite known number of approved mobile devices are allowed to access the store's wireless network, the best security method to implement on the access points would be MAC filtering whereby only those specifically allowed devices would be able to connect.

Objective:

Network Fundamentals

Sub-Objective:

Describe wireless principles

References:

Global Sign > 12 Best Practices for Wireless Network Security

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Question #4 of 10

Question ID: 1704754

Which of the following is FALSE with respect to spine-leaf topology?

A) The spine switches are connected to the backbone.

B) **The spine and leaf layer are connected in a full mesh topology.**

C) The leaf switches are connected to the end devices.

D) It has two layers.

Explanation

The spine switches are not connected to the backbone. They ARE the backbone and interconnect all the leaf switches.

The leaf switches are connected to the end devices and to the spine switches.

There are two layers in the topology: the spine layer, and the leaf layer.

The spine and leaf layer are connected in a full mesh topology, providing redundancy.

Objective:

Network Fundamentals

Sub-Objective:

Describe characteristics of network topology architectures

References:

Geeks for Geeks > Spine-Leaf Architecture

Question #5 of 10

Question ID: 1703575

Which of the following are types of Internet Protocol version 6 (IPv6) addresses? (Choose three.)

A) Dual-cast

B) Multicast

C) Unicast

D) Anycast

E) Broadcast

Explanation

Unicast, multicast, and anycast are types of IPv6 addresses.

Unicast addresses are used to define a single destination interface. A packet sent to a unicast address is delivered to the specific interface.

Multicast addresses are used to define a group of hosts. When a packet is sent to a multicast address, it is delivered to all the hosts identified by that address. Multicast addresses begin with the prefix FF00::/8 and the second octet identifies the range over which the multicast address is propagated. Some special case IPv6 multicast addresses are:

- FF01:0:0:0:0:0:0:1 – Indicates all-nodes address for interface-local scope.
- FF02:0:0:0:0:0:0:2 – Indicates all-routers address for link-local.

Anycast addresses are used to identify a set of devices. These addresses are also assigned to more than one interface belonging to different nodes. A packet sent to an anycast address is delivered to just one of the interfaces, based on which one is closest. For example, if an anycast address is assigned to a set of routers, one in India and another in the U.S., the users in the U.S. will be routed to U.S. routers and the users in India will be routed to a server located in India.

The broadcast option is incorrect because these types of addresses are not supported by IPv6. Broadcast functionality is provided by multicast addressing.

The dual-cast option is incorrect because this is not a valid Cisco address type.

Objective:
Network Fundamentals

Sub-Objective:
Describe IPv6 address types

References:

Cisco Press > Articles > IPv6 Address Representation and Address Types

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Question #6 of 10

Question ID: 1704762

What is the term used for the Ethernet communication mechanism by which hosts can send and receive data simultaneously?

A) Half-duplex

B) **Full-duplex**

C) Duplex

D) Multiplex

Explanation

Full-duplex communication occurs when workstations can send and receive data simultaneously. To support full-duplex communication, both communicating hosts should be configured to transmit in full-duplex mode. With the use of full-duplex communication, the bandwidth can effectively be doubled. Hubs are not capable of handling full-duplex communication, and you need a dedicated switch port to allow full-duplex communication.

It is important that the switch and the device connected to the switch have the same duplex and speed settings, or there will be intermittent connectivity and loss of connection. To verify the duplex and speed settings on a switch, execute the **show interfaces** command specifying the interface, and the setting can be verified (as shown in line 8 in the output below):

```
switch# show interface fastethernet 0/3
Fast Ethernet 0/3 is down, line protocol is down (not connect)
Hardware is Fast Ethernet, address is 00e0.1e3e.2a02
MTU 1500 bytes, BW 10000 Kbit, DLY 100 usec, rely 1/255, tx load
1/255, rxload 1/255
Encapsulation ARPA, loopback not set,
Keepalive set (10 sec)
Half-duplex, 100Mb/s
ARP type: ARPA, ARP Timeout 04:00:00
```

From the output above, it can be seen that the switch interface is set for half-duplex and the speed is set for 100 Mbps. This means that if the host connected to this switch port is set differently, for example set to 1 Gbps because it has a 1 Gb NIC, the host and the switch interface will not communicate, and the host will not be able to connect to the network.

Half-duplex is the term used for the Ethernet communication mechanism when hosts can send or receive data, but not simultaneously.

Multiplex is the term used when multiple signals are combined to be transferred via one signal.

Duplex implies that there are two communication paths. However, the term does not specify the required functionality, which is full-duplex.

Objective:

Network Fundamentals

Sub-Objective:

Identify interface and cable issues (collisions, errors, mismatch duplex, and/or speed)

References:

Cisco > Support > Technology Support > Ethernet > Troubleshooting TechNotes > Configure and Verify Ethernet 10/100/1000Mb Half/Full Duplex Auto-Negotiation > Document ID: 10561

Question #7 of 10

Question ID: 1703531

RouterA and RouterB, which connect two locations, are unable to communicate. You run the **show running-configuration** command on both router interfaces, RouterA and RouterB. The following is a partial output:

```
routerA#show running-config
interface Serial0
description Router_A
ip address 192.10.191.2 255.255.255.0
encapsulation ppp
no ip mroute-cache
clockrate 64000
routerB#show running-config
interface Serial1
description Router_B
ip address 192.10.192.1 255.255.255.0
encapsulation ppp
no ip mroute-cache
clockrate 64000
```

Based on the information given in the output, what are the likely causes of the problem? (Choose two.)

A) **At least one of the routers must have the ip mroute-cache command enabled.**

B) The IP address defined is incorrect.

C) The Layer 2 framing is misconfigured.

D) **Both routers cannot have an identical clock rate.**

E) Both routers cannot have a clock rate defined.

Explanation

Two possible causes of the problem are that the IP addresses are incorrect as defined, or that both routers have a defined clock rate. The IP addresses on the routers are in different subnets. The IP addresses need to be changed to fall in the same subnet.

Both routers cannot have a clock rate configured. Only routers with a DCE cable connected should have a clock rate, which provides synchronization to the router connected to the DTE cable. In a point-to-point serial connection, the DCE cable connects to the DTE cable, providing a communication path between the two routers. If both computers have a clock rate configured, the routers will not communicate.

A matching clock rate is not the problem. The clock rates between two routers should match. The router connected to the DCE cable will provide the clock rate to the router connected to the DTE cable, resulting in matching clock rates.

Layer 2 encapsulation refers to the Data Link protocol used on the link. In this case, the protocol is Point-to-Point Protocol (PPP), which is configured correctly on both ends as indicated by the matching **encapsulation ppp** statements in the output. The connection would be prevented from working if one of the routers were missing this setting (which would be indicated by the absence of the **encapsulation ppp** statement in its output), or if a different Layer 2 encapsulation type were configured, such as High-Level Data Link Control (HDLC).

The **ip mroute-cache** command is used to fast-switch multicast packets and would not cause the problem in this scenario.

Objective:
Network Fundamentals

Sub-Objective:
Identify interface and cable issues (collisions, errors, mismatch duplex, and/or speed)

References:

Cisco > Cisco IOS XE 16 > Command References > Cisco IOS Interface and Hardware Component Command Reference > clock rate through cut-through > clock rate

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Question #8 of 10

Question ID: 1703539

Refer to the following partial output of the **show interfaces** command:

```
Serial 0 is down, line protocol is down
Hardware is MCI Ethernet, address is 0000.0c00.750c (bia 0000.0c00.750c)
Internet address is 134.108.28.8, subnet mask is 255.255.255.0
MTU 1500 bytes, BW 10000 Kbit, DLY 100000 usec, rely 255/255, load 1/255
Encapsulation ARPA, loopback not set, keepalive set (10 sec)
ARP type: ARPA, ARP Timeout 4:00:00
```

```
Last input 0:00:00, output 0:00:00, output hang never
Last clearing of "show interface" counters 0:00:00
Output queue 0/40, 0 drops; input queue 0/75, 0 drops
Five minute input rate 0 bits/sec, 0 packets/sec
Five minute output rate 2000 bits/sec, 4 packets/sec
1127576 packets input, 447251251 bytes, 0 no buffer
Received 354125 broadcasts, 0 runts, 0 giants, 57186* throttles
0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
5332142 packets output, 496316039 bytes, 0 underruns
0 output errors, 432 collisions, 0 interface resets, 0 restarts
<<output omitted>>
```

What troubleshooting steps should you perform to resolve the problem depicted in the output? (Choose two.)

A) Reset the equipment.

B) **Check the router configuration.**

C) **Check the router configuration for the shutdown interface command.**

D) Check the cable connections.

Explanation

You should check the cable connections and reset the equipment to troubleshoot the problem depicted in the output. The `Serial 0 is down, line protocol is down` message indicates that there is no carrier detect (CD) signal sensed by the router. This problem might be due to incorrect cabling or hardware failure.

A complete list of the possible troubleshooting steps that should be performed to resolve this issue include:

- Checking the cable connections.
- Resetting the equipment.
- Checking the CD LED on the CSU/DSU.
- Reporting the issue to the leased-line provider.
- Replacing the faulty equipment.

The router configuration is not an issue in this scenario because both serial 0 and line protocol are down, indicating a problem in the physical layer. Configuration issues, such as an incorrect IP address, would be indicated in the second section of the output (`line protocol is up/down`). The second section, regardless of whether it says up or down is meaningless when the first section indicates a problem.

You would not check the router configuration for the **shutdown interface** command. When an interface has been manually shut down with this command, it will be indicated in the output as `Serial 0 is administratively down, line protocol is down`.

Objective:

Network Fundamentals

Sub-Objective:

Identify interface and cable issues (collisions, errors, mismatch duplex, and/or speed)

References:

Cisco > Basic Router Commands and Tasks

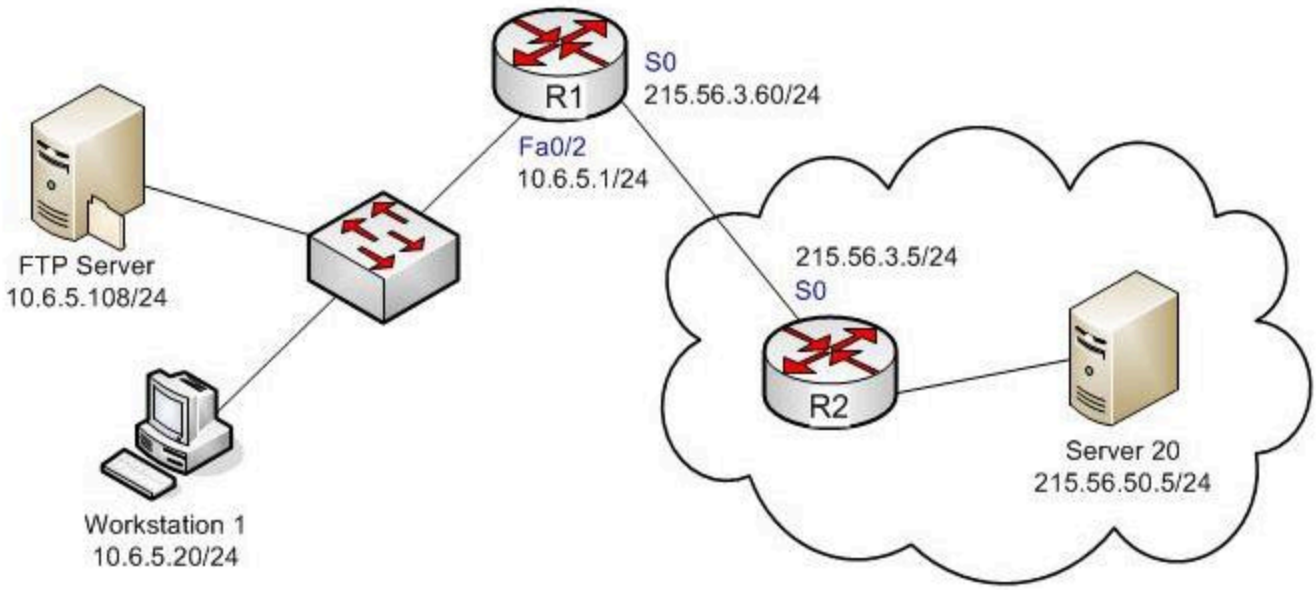
Cisco > Product Support > Cisco IOS and NX-OS Software > Cisco IOS Interface and Hardware Component Command Reference > show interfaces

Cisco > Support > Technology Support > T1/E1 & T3/E3 > Troubleshooting TechNotes > Troubleshooting Serial Line Problems > Document ID: 14149

Question #9 of 10

Question ID: 1703565

To enable users to access the FTP server from the Internet, which service must be operational on which device?



A) DHCP on R1

B) NAT on R2

C) NAT on R1

D) **VTP on the switch connected to the FTP server**

Explanation

Network Address Translation (NAT) must be operational on the R1 router. The computers on the LAN side of R1, including the FTP server, are using a private IP address range (10.6.5.0/24). To allow access to the FTP server from the Internet, the private IP address of the FTP server (10.6.5.108/24) must be converted to a public IP address. NAT can perform this translation and should be operational on the router that connects the LAN where the FTP server is located to the Internet, which in this case is R1.

NAT does not need to be running on R2. Both of the interfaces on R2 are using public IP addresses, so no translation is necessary. Moreover, configuring R2 would not help users connect to the FTP server since R1 is the router that connects the LAN where the FTP server is located to the Internet.

Making DHCP operational on R1 would not help. DHCP provides address configurations to both local and remote clients but cannot perform the translation required in this scenario.

Making VLAN Trunking Protocol (VTP) operational on the switch would not provide any translation. This protocol is used to communicate VLAN information among multiple switches.

Objective:
Network Fundamentals

Sub-Objective:
Describe private IPv4 addressing

References:

What Is Network Address Translation (NAT)?

Question #10 of 10

Question ID: 1703532

A switch is powered up and the system LED is amber. Which of the following describes this situation?

A) **The switch is performing normally.**

B) Utilization level is high.

C) The switch is malfunctioning.

D) There is a security violation on a switch port.

Explanation

The system LED indicates the overall health of the switch. The LED should turn solid green after a successful Power On Self-Test (POST). An amber system LED indicates that there is a system-wide failure in the switch.

High utilization will not cause the system LED to turn amber.

An amber system LED indicates a general switch malfunction. It does not indicate that the switch is performing normally.

Port security violations will not cause the system LED to be amber. The system LED is used to identify the overall health of the switch.

Objective:
Network Fundamentals

Sub-Objective:
Identify interface and cable issues (collisions, errors, mismatch duplex, and/or speed)

References:

Cisco > Product Support > Switches > Catalyst 2960 Switch Hardware Installation Guide > Chapter: Product Overview > LEDs

Hewlett Packard Enterprise > Cisco Catalyst 2960 Series Switch - Monitoring and Troubleshooting Using LEDs